In-Orbit Test of the Polarized Scanning Atmospheric Corrector (PSAC) Onboard Chinese Environmental Protection and Disaster Monitoring Satellite Constellation HJ-2 A/B

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***Abstract*— As the successors of the overdue HuanjingJianzai-1 (HJ-1) satellites and new members in Chinese Environmental Protection and Disaster Monitoring Satellite Constellation, the first two of HuanjingJianZai-2 series satellites (HJ-2 A/B) have been launched on September 27, 2020. Each satellite carries four sensors, including the polarized scanning atmospheric corrector (PSAC), the charge-coupled device (CCD) camera, the hyperspectral imager (HSI), and the infrared spectroradiometer (IRS). Among them, PSAC is mainly used for the monitoring of atmospheric parameters to provide data support for atmospheric environmental monitoring and atmospheric correction of data from other sensors. To test the in-orbit performance of PSAC, we develop the "day-1" aerosol and water vapor retrieval algorithms. The preliminary validation results based on ground-based observations show that the aerosol optical depth (AOD) and columnar water vapor (CWV) datasets developed based on PSAC data have high accuracy and can effectively characterize the temporal trends of AOD and CWV. The accuracy of PSAC AOD dataset is better than the expected error (EE) $\pm(0.05 + 0.2 * \mathrm{AOD}_{\mathrm{AERONET}})$, and the accuracy of PSAC CWV dataset is better than the EE $\pm(0.05 + 0.15 * \mathrm{CWV}_{\mathrm{AERONET}})$. To eliminate the negative impact of the atmosphere on CCD data and expand its application range, aerosol and water vapor data developed based on PSAC are used for atmospheric correction of CCD data. Compared with L1 CCD data, the texture details and clarity of CCD data after atmospheric correction have been significantly improved.**

***Index Terms*— Atmospheric correction, HuanjingJianzai-2 (HJ-2) satellite, in-orbit test, polarized scanning atmospheric corrector (PSAC), remote sensing.**

I. Introduction

ENVIRONMENT is the basis for human survival, and Environmental Satellites (ENVISAT) are the only space coverage means to monitor and evaluate the global and regional environment. ENVISAT can carry out comprehensive monitoring of air, water, soil, and ecology, with the Earth Observing System (EOS) of National Aeronautics and Space Administration (NASA) [1]–[4] and the ENVISAT of European Space Agency (ESA) [5]–[8] as representatives having an important impact on the world. The new generation of ENVISAT takes higher spatial resolution, more accurate measurement, and more dimensional information as the developing direction. Among them, atmospheric sensors are mainly responsible for the detection of atmospheric components, such as aerosol, cloud, water vapor, and so on. These components are closely related to meteorological climate, air pollution, climate change, etc., [9]–[11], and are the focus of attention on the atmospheric environment. Aerosols and clouds are particle-like substances in the atmosphere, their composition and morphology are complex and changeable. Accurate description of their characteristics and effects requires multidimensional complex parameters (and models) [12], [13], which poses a great challenge to remote sensing detection.

Internationally, the Aerosol Polarimetry Sensor (APS) [14] was developed by the NASA, but it has not been successful due to the failure of glory satellite launch. Other comparable instruments include the Polarization and Directionality of the

Manuscript received January 7, 2022; revised April 9, 2022; accepted May 13, 2022. Date of publication May 23, 2022; date of current version June 6, 2022. This work was supported in part by the National Outstanding Youth Foundation of China under Grant 41925019, in part by the National Natural Science Foundation of China under Grant 41871269 and Grant 41701407, and in part by the K. C. Wong Education Foundation under Grant GJTD-2018-15. *(Corresponding authors: Weizhen Hou; Zhenhai Liu.)*

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Digital Object Identifier 10.1109/TGRS.2022.3176978

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Earth's Reflectances (POLDER) series sensors [15] developed by French and the Directional Polarimetric Camera (DPC) [16] onboard Gaofen-5 (GF-5) and GF-5 (02) satellites in Chinese High-Resolution Earth Observation System (CHEOS) program. The main advantage of POLDER and DPC type sensors is to obtain multispectral, multiangle, intensity, and polarization image information at the same time. However, the spectral wavelength coverage is relatively limited due to the instrument mechanism, such as the lack of ultraviolet (UV) and short-wave infrared (SWIR) wavelength bands. Besides, it is difficult to achieve high accuracy in calibration, which affects the use of fine polarization information. In order to solve these key problems, the multiviewing multichannel multipolarization imager (3MI) [17] has been developed by the European Organization for the Exploitation of Meteorological Satellites (EUMETSAT), which will be launched after the year of 2022.

The first two of Chinese HuanjingJianzai-2 (HJ-2) series satellites, also called HJ-2 A/B, were successfully launched on September 27, 2020, working as the successor of the overdue HuanjingJianzai-1 (HJ-1) A/B satellites in Chinese Environmental Protection and Disaster Monitoring Satellite Constellation. Correspondingly, better remote sensing performance, image quality, imaging performance, and satellite reliability could be provided by HJ-2 A/B satellites. There are four main sensors onboard HJ-2 A/B satellites, including the polarized scanning atmospheric corrector (PSAC), the charge-coupled device (CCD) camera, the hyperspectral imager (HSI) and the infrared spectroradiometer (IRS) [18]. Among them, PSAC is a new polarimeter with nine polarized channels from 410 to 2250 nm, which is mainly used to detect the key atmospheric parameters such as aerosols, clouds and water vapor with high precision, and further to provide the atmospheric properties for synchronous atmospheric correction of the main sensors onboard the same HJ-2 satellites. With the cross-track scanning mode of PSAC, the total scanning angle range of PSAC can reach 65° from −32.5° to 32.5° in nadir observation, and then a wide-swath coverage larger than 800 km with the spatial resolution of about 6 km can be obtained. Correspondingly, the multispectral polarized measurements of PSAC can fully cover other three main sensors integrated on the same satellite platform, aiming to provide the atmospheric properties for synchronous atmospheric correction of the multispectral and hyperspectral images of other main sensors.

Based on the "day-1" retrieval algorithm for the in-orbit test of HJ-2 A/B satellites, the preliminary aerosol optical depth (AOD) and columnar water vapor (CWV) retrieval results are derived from the cloud-free pixels of PSAC, and the synchronous atmospheric correction of CCD image is also carried out. In this study, the HJ-2 A/B satellites and PSAC instrument are briefly introduced in Section II. Then, the retrievals and validations of AOD and CWV are investigated in Section III. After that, the synchronous atmospheric corrections of multispectral images are presented in Section IV. Subsequently, we give supplemented discussion in Section V. Finally, the conclusions of the work are summarized in Section VI.

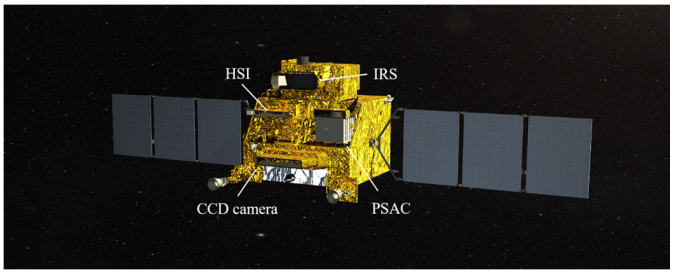


Fig. 1. Four sensors onboard HJ-2 A/B satellites.

II. HJ-2 A/B Satellites and PSAC Instrument

A. HJ-2 A/B Satellites

The launched Chinese HJ-2 series satellites run in a sun-synchronous orbit, and the local time of descending node is about 10:30 A.M. There are two satellites with the same configuration, HJ-2A and HJ-2B, arranged in 180° phase to obtain a higher time resolution. Just as shown in Fig. 1, there are four typical sensors integrated in the platform of HJ-2 A/B satellites, including CCD camera, HSI, IRS, and PSAC, respectively, [18]. Correspondingly, Table I lists the detailed instrumental parameters of the four main sensors onboard HJ-2 A/B satellites, and Table II lists the specific definition of PSAC and HJ-2 CCD data products at different levels.

Among these four instruments, PSAC is a new cross-track scanning polarimeter developed by China, which has nine polarized channels from near-UV to SWIR and centered in 410, 443, 555, 670, 865, 910, 1380, 1610, and 2250 nm with the polarized angle 0°, 45°, 90°, and 135°, respectively. With a large scanning angle (65°) and a wide-swath coverage, a scene of PSAC multispectral image onboard HJ-2 satellites is about 720 km × 880 km with the 154 × 130 pixels and the nadir spatial resolution of about 6 km. Compared with the CCD sensors onboard HJ-1 A/B satellites, the red edge band (0.69–0.73 μm) has been added to indicate the growth status of green plants, with the number of multispectral wavelength bands increasing from 4 to 5 and the spatial resolution increasing from 30 to 16 m for CCD/HJ-2. Compared with the HSI/HJ-1A, the hyperspectral range of HIS/HJ-2 is extended from visible and near-infrared (VIS-NIR) bands (0.45–0.95 μm) to SWIR bands (0.45–0.92 μm, 0.90–2.5 μm). In addition, the number of bands increases from 110 to 220, the spatial resolution improves from 100 to 48 m for VIS-NIR bands and 96 m for NIR-SWIR bands, and the swath width extends from 50 to 96 km. Compared with the IRS/HJ-1B, the number of used wavelength bands of IRS/HJ-2 is expanded from 4 to 9, the spatial resolution is improved from 150 m (0.75–1.1, 1.55–1.75, 3.5–3.9 μm) and 300 m (10.5–12.5 μm) to 48 m (B1–B5) and 96 m (B6–B9), respectively, which can also maintain a large swath with about 720 km as CCD cameras.

B. Imaging Principle and Data Preprocessing of PSAC/HJ-2

PSAC is a cross-track scanning polarimeter with nine polarized channels from near-UV to SWIR, centered in 410, 443, 555, 670, 865, 910, 1380, 1610, and 2250 nm, respectively.

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TABLE I
INSTRUMENTAL PARAMETERS OF FOUR MAIN SENSORS ONBOARD HJ-2 A/B SATELLITES

Parameter	Four sensors onboard HJ-2 A/B satellites			
	PSAC	**CCD**	**HSI**	**IRS**
Nadir pixel resolution	6 km	16 m	48 m (0.45-0.92 μm) 96 m (0.90-2.5 μm)	48 m (B1-B5) 96 m (B6-B9)
Swath width	800 km	800 km	96 km	720 km
Wavelength bands	0.40-0.42 μm 0.43-0.45 μm 0.55-0.57 μm 0.66-0.68 μm 0.85-0.89 μm 0.90-0.92 μm 1.36-1.40 μm 1.58-1.64 μm 2.21-2.29 μm	0.45-0.52 μm 0.52-0.59 μm 0.63-0.69 μm 0.69-0.73 μm 0.77-0.89 μm	0.45-0.92 μm (Δλ=5 nm) 0.90-2.5 μm (Δλ=14 nm)	0.63-0.69 μm 0.73-0.77 μm 0.78-0.90 μm 1.19-1.29 μm 1.55-1.68 μm 2.08-2.35 μm 3.50-4.80 μm 10.5-11.4 μm 11.5-12.5 μm
Laboratory calibration accuracy	Radiance: ≤7% DOLP: 0.005	Absolute: ≤7% Relative: ≤3%	Absolute: ≤7% Relative: ≤3%	B1-B6: absolute ≤7%, relative ≤3%

TABLE II
SPECIFIC DEFINITION OF PSAC AND HJ-2 CCD DATA PRODUCTS AT DIFFERENT LEVELS

Payload	Level	Content
PSAC	L0	Raw data after scene splitting
	L1	9 bands Stokes vector elements (I, Q and U) normalized radiance data with the following information attached: Geolocation QC flags
	L2	Atmospheric parameters retrieved based on L1 product: Aerosol optical depth (AOD) Columnar water vapor (CWV) Cloud detection
CCD	L0	Raw data after scene splitting
	L1A	The original pixel after relative radiometric correction, with the following information attached: Absolute radiometric calibration coefficient Rational polynomial coefficients (RPC)
	L1B	Surface reflectance after atmospheric correction based on L1A product, with the following information attached: Rational polynomial coefficients (RPC)

With a large scanning angle (±32.5° from nadir) and a wide-swath coverage with the spatial resolution of about 6 km at nadir, a scene of PSAC image is about 720 km × 880 km with the 154 × 130 pixels. The prototype of PSAC is shown in Fig. 2.

Its measurement principle is the same as that of American APS/Glory (Aerosol Polarimetry Sensor/Glory Mission) [19] and the Chinese Particulate Observing Scanning Polarimeter (POSP) [20], [21] that is mounted on the GaoFen-5 (02) and the atmospheric environmental monitoring (DQ-1) satellites. They adopt a multispectral, split-aperture, split-amplitude optical system for simultaneous polarization measurement. The basic principle of PSAC polarization measurement is based on the fact that two Wollaston prisms used in its optical system are equivalent to four polarization analyzers with polarization azimuths of 0°, 90°, 45°, and 135°, and the light intensity detected by the four corresponding detectors is I_0, I_{90}, I_{45}, and I_{135}. Given the target's Stokes vector $[I\ Q\ U\ V]^T$, the measurement equation can be expressed as

$$\begin{bmatrix} I \\ Q \\ U \\ V \end{bmatrix} = \begin{bmatrix} I_0 + I_{90} \\ I_0 - I_{90} \\ I_{45} - I_{135} \\ 0 \end{bmatrix}. \tag{1}$$

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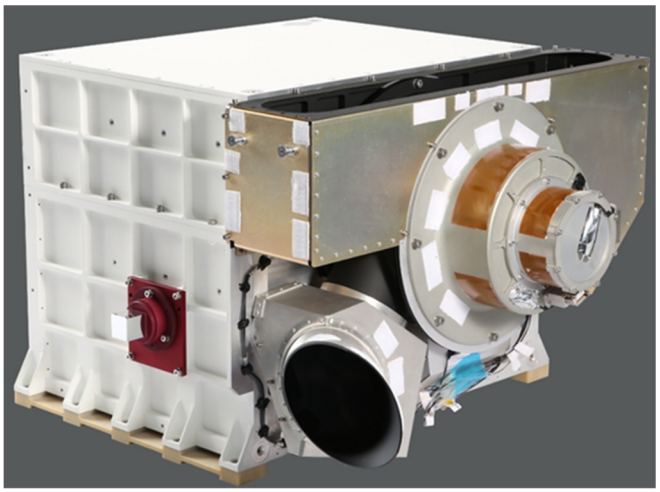


Fig. 2. Prototype of PSAC onboard HJ-2 A/B satellites.

For naturally illuminated scenes on the Earth, the circular polarization component V is at least two orders of magnitude smaller than the linear polarization components Q and U and thus can be ignored in atmospheric detection [22]. Hence, the degree of linear polarization (DoLP) and the azimuth angle of linear polarization (AoLP) can be written as

$$\begin{cases} \text{DoLP} = \dfrac{\sqrt{Q^2+U^2}}{I} \\ \text{AoLP} = \dfrac{1}{2}\arctan\left(\dfrac{U}{Q}\right). \end{cases} \tag{2}$$

PSAC is equipped with nine spectral bands for atmospheric parameter retrieval and application requirements, and a total of three pairs of six independent optical paths are used for detection. Each pair of optical paths uses Wollaston prisms for polarization beam splitting, and a combination of dichroic and optical filters are used to realize the splitting of three spectral bands. In order to ensure the long-term in-orbit detection accuracy, the PSAC is equipped with four kinds of onboard calibrators, including the solar reference assembly (SRA), the dark reference assembly (DRA), the polarized reference assembly (PRA) and the unpolarized reference assembly (URA). The DRA is used for dark current correction and the SRA is used for onboard absolute radiometric calibration, while the PRA and URA are used for onboard polarimetric calibration. These calibrators are arranged around the orthogonal scanning mirror and can be observed during each scan.

The PSAC data products are organized into three levels: raw data as level 0 (L0) data, sensor data after preprocessing as level 1(L1) data, and environmental data retrieved based on L1 as level 2 (L2) data. PSAC data preprocessing refers to the process from L0 to L1. The L0 raw data of the satellite download data received by the ground data processing system after unpacking and split-scene processing includes: 1) all the 36 channels remote sensing data observed by PSAC; 2) the remote sensing data related to the state of the satellite remote sensor, such as voltages, currents, temperatures, motor speed and status parameters, etc.; and 3) Satellite platform telemetry data, including GPS time, satellite position, attitude data, and so on.

L1 is generated from L0 data and instrument characteristic data. Products include remote sensor data quality check (QC), polarization analysis of the observations, and geolocation of the scene samples. The L0 to L1 data preprocessing flow is as follows.

1) The data QC of PSAC is to ensure that the instrument is within operating limits and the L0 data is valid. The specific items include checking the health status of the remote sensor, the validity of the scientific data, and the validity of the key parameters of the platform, etc., and abnormal conditions such as channels failure and parameters exceeding the normal range will be flagged.
2) Data precorrection of the original data of the remote sensor, including the calculation and correction of dark counts for each channel, channel drift correction, and temperature correction for the shortwave infrared channels, and so on.
3) Calculation of Stokes vector elements (I, Q, U) of the scene, the radiometric and polarimetric calibration parameters involved include absolute radiation calibration coefficients, errors in orientation of the Wollaston prism, Wollaston prism extinction ratios, intertelescope/intratelescope calibration factors, instrumental polarization, the polarization scaling factors, etc., as shown in (3) and (4), the scene data are converted to normalized radiance data (I_N, Q_N, U_N) with the solar geometry/irradiance information

$$\begin{bmatrix} I_N \\ Q_N \\ U_N \end{bmatrix} = \frac{\pi d^2}{\bar{E}_0} \cdot \begin{bmatrix} I \\ Q \\ U \end{bmatrix} \tag{3}$$

$$\bar{E}_0 = \frac{\int_{\Delta f} E_0(\lambda) \cdot \text{SRF}(\lambda) d\lambda}{\int_{\Delta f} \text{SRF}(\lambda) d\lambda} \tag{4}$$

where d is the astronomical distance between the sun and the earth at the time of observation, $E_0(\lambda)$ is the solar spectral irradiance at the top of the earth's atmosphere at the average distance between the sun and the earth, Δf is the spectral bandwidth, and $\text{SRF}(\lambda)$ is the relative spectral response function (SRF) of PSAC.
4) The geolocation of each ground sampling point is determined by the given satellite position, time, attitude angles, sampling timing, sensor scanning angle and orientation.

III. Retrieval Algorithm for Aerosol and Water Vapor and Preliminary Results

The flowchart of aerosol and water vapor retrieval is shown in Fig. 3.

A. Retrieval Algorithm

1) Cloud Mask Algorithm: The algorithms for retrieving aerosol and water vapor used in this work are only applicable to cloud-free pixels. In order to avoid the occurrence of retrieval results with large errors caused by cloudy pixels,

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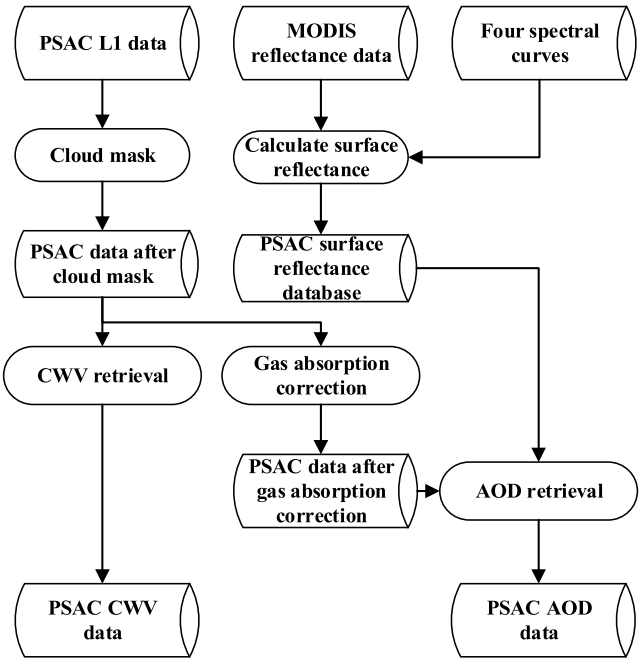


Fig. 3. Flowchart of AOD and CWV retrieval.

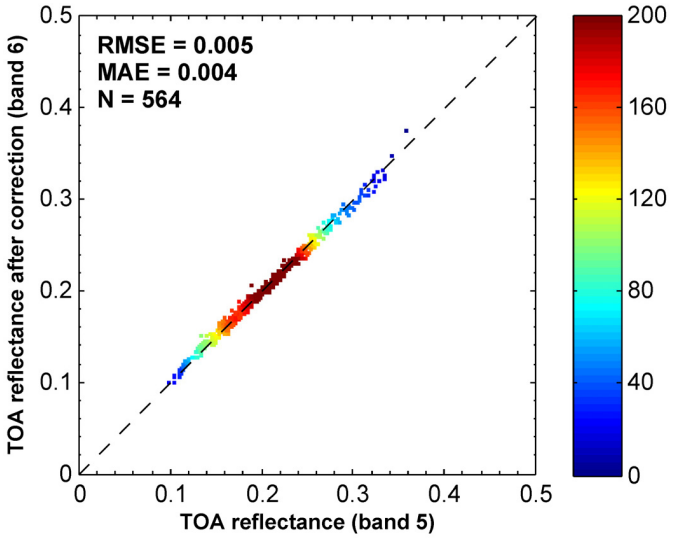


Fig. 4. Scatter plot of the matching results between the TOA reflectance of band six after gas absorption correction and the TOA reflectance of band 5.

three methods based on different principles are used together to identify cloudy pixels. If a pixel is judged as a cloudy pixel by any method, the pixel will be rejected as a cloudy pixel so as to eliminate the cloudy pixel as much as possible. The details of the above three methods are as follows.

1) The top-of-atmosphere (TOA) reflectance of band 5 at 865 nm and band 6 at 910 nm is used for cloud identification. Due to their relatively long and close wavelengths, the differences between path radiation and atmospheric transmittances of bands 5 and 6 can be ignored. In addition, the difference in surface reflectance between these two bands is very small. For the above reasons, it can be considered that the TOA reflectance of band 5 is equal to that of band 6 without considering atmosphere absorption. In addition, both the absorption effect of the atmosphere on band 5 and the absorption effects of gases other than water vapor on band 6 can be neglected. Therefore, the TOA reflectance of these two bands can be regarded as approximately equal without considering the absorption of water vapor on the signal received by the satellite [23], [24].
   The scatter plot of the matching results between the TOA reflectance of band 6 after gas absorption correction using CWV data derived from Aerosol Robotic Network (AERONET) and the TOA reflectance of band 5 are shown in Fig. 4. Note that AERONET is a ground-based observation network that can provide AOD and CWV data with extremely high quality [25]. Here the mean absolute error (MAE) and root mean square error (RMSE) are used to assess the difference between the TOA reflectance of band 6 after gas absorption correction and the TOA reflectance of band 5

$$\mathrm{RMSE} = \sqrt{\sum_{i=1}^{N} \left(P_i - P_i'\right)^2 / N} \tag{5}$$

$$\mathrm{MAE} = \sum_{i=1}^{N} \left|P_i - P_i'\right| / N \tag{6}$$

where N is the number of samples, P_i is the value of the ith TOA reflectance of band 6 after gas absorption correction, and P_i' is the value of the TOA reflectance of band 5 corresponding to P_i.
As shown in Fig. 4, both the RMSE (0.005) and MAE (0.004) used to assess the difference between the TOA reflectance of band 6 after gas absorption correction and the TOA reflectance of band 5 are very small. This means that the TOA reflectance of band 6 without considering water vapor absorption and the TOA reflectance of band 5 can indeed be considered equal. For the actual atmosphere that is not affected by clouds, the signal of band 6 is significantly reduced due to water vapor absorption, and the signal of band 5 is hardly affected by atmospheric absorption, so the TOA reflectance of band 6 should be smaller than that of band 5. Therefore, the pixels whose TOA reflectance of band 6 is larger than that of band 5 will be identified as cloudy pixels.

2) The TOA reflectance of band 2 at 443 nm and band 7 at 1380 nm is used to identify cloudy pixels. If the TOA reflectance of band 7 is greater than 0.02 or the TOA reflectance of band 2 is greater than 0.4, the pixel will be regarded as cloudy pixel. Note that the above thresholds are determined by analyzing a large number of PSAC data. This method can effectively identify thick clouds, and similar methods have been successfully applied to moderate-resolution imaging spectroradiometer (MODIS) and Advanced Himawari Imager (AHI) with similar wavebands [26], [27].
3) The spatial variability of TOA reflectance of band 2 and band 7 is used to identify clouds [26]. For a pixel to be detected, the standard deviation (STD) of TOA reflectance of each band between the pixel and its nearest 8 pixels will be calculated first. The method of calculating STD is shown in (7). Then, if the calculated

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STD of band 7 is greater than 0.005 or the STD of band 2 is greater than 0.01, the pixel will be regarded as a cloudy pixel and removed. Note that the threshold used here is also determined by analyzing a large number of data. This method can effectively identify thin clouds and cloud edges

$$\mathrm{STD} = \sqrt{\frac{\sum_{i=1}^{9}\left(\rho_{\mathrm{TOA}}^{i} - \rho_{\mathrm{TOA}}^{\mathrm{mean}}\right)^2}{9}} \tag{7}$$

where ρ_{TOA}^{i} is the TOA reflectance with different serial number, $\rho_{\mathrm{TOA}}^{\mathrm{mean}}$ is the average value of all 9 TOA reflectance participating in the calculation, and STD represents the calculated standard deviation.

2) Water Vapor Retrieval Algorithm: A total of two NIR bands at 865 and 910 nm are used for water vapor retrieval. As mentioned above, since the TOA reflectance of the two bands can be regarded as approximately equal without considering atmosphere absorption, and both the absorption effect of the atmosphere on band 5 and the absorption effects of gases other than water vapor on band 6 can be neglected, the water vapor absorption transmittance of band 6 can be calculated according to the following equation [23], [24], [28]:

$$T_{910\,\mathrm{nm}} = \rho_{\mathrm{TOA}}^{910\,\mathrm{nm}} / \rho_{\mathrm{TOA}}^{865\,\mathrm{nm}} \tag{8}$$

where $\rho_{\mathrm{TOA}}^{865\,\mathrm{nm}}$ is the TOA reflectance of band 5, $\rho_{\mathrm{TOA}}^{910\,\mathrm{nm}}$ is the TOA reflectance of band 6, and $T_{910\,\mathrm{nm}}$ is the water vapor absorption transmittance of band 6.

Since the SRF of band 6 is known, we can use the Second Simulation of the Satellite Signal in the solar spectrum-vector (6SV) radiative transfer model [29] to calculate the water vapor absorption transmittance of the band at any water vapor content. The above is the basis of water vapor retrieval. The detailed water vapor retrieval procedure is as follows. For a specific pixel, we can first calculate the water vapor absorption transmittance of band 6 (referred to as T1) for the pixel by using the TOA reflectance of band 5 and band6 according to (8). Then, we can calculate 61 water vapor absorption transmittances of band 6 under different water vapor contents (CWV range: 0–6 cm, interval: 0.1 cm) by using the 6SV radiative transfer model based on the observation geometry of the pixel. After that, we can find two adjacent water vapor absorption transmittances from these 61 water vapor absorption transmittances, one of which (referred to as T2) is not less than T1 and the other (referred to as T3) is not greater than T1. Finally, CWV retrieval can be achieved by linear interpolation based on T1, T2, T3, water vapor content corresponding to T2 (referred to as V_{CWV2}) and water vapor content corresponding to T3 (referred to as V_{CWV3})

$$V_{\mathrm{CWV}} = V_{\mathrm{CWV2}} + \frac{(V_{\mathrm{CWV3}} - V_{\mathrm{CWV2}}) * (\mathrm{T1} - \mathrm{T2})}{(\mathrm{T3} - \mathrm{T2})} \tag{9}$$

where V_{CWV} is the value of the CWV retrieval result.

3) AOD Retrieval Algorithm: In this work, data from four bands (443, 555, 670, and 865 nm) of PSAC are used for aerosol retrieval. For aerosol retrieval, it can be assumed that the ratios between surface reflectance of different bands do not change within one month [30], [31]. The commonly used method to construct the ratio of surface reflectance of different bands is as follows. For a specific location, first select the subdark pixels of different bands at this location, then perform atmospheric correction on the subdark pixels under the assumption that the subdark pixels are not affected by aerosols to obtain the surface reflectance of different bands of the subdark pixels, and finally calculate the ratio of the surface reflectance of different bands at the location [31], [32]. The above method requires the use of long time series of observation data. However, PSAC is still in the stage of in-orbit testing and the available PSAC data are not sufficient for the construction of ratios of surface reflectance of different bands using the method.

To solve the problem, the ratios between surface reflectance of different bands of PSAC for each month are constructed by spectral reconstruction technique based on MODIS surface reflectance data released by NASA [33]. First, a total of four spectral curves are extracted from spectral library containing various typical features based on the principle of nonnegative matrix factorization (NMF) method [34]. Then, the weights of each spectral curve at different positions can be calculated with MODIS surface reflectance as a constraint [34], [35]. After obtaining the weights of the four spectral curves at different locations, we can calculate the surface reflectance of different bands of PSAC at different locations based on the spectral curves and corresponding weights. Then we can obtain the ratios of surface reflectance of band 5 to that of other three bands. After that, the TOA reflectance of the other three bands at different aerosol contents can be simulated by using the calculated ratio and the TOA reflectance of band 5 according to the following equation [36]:

$$\rho_{\mathrm{TOA}}^{\lambda}(\theta_s, \theta_v, \varphi) = \rho_0^{\lambda}(\theta_s, \theta_v, \varphi) + \frac{T_{\downarrow}^{\lambda}(\theta_s) T_{\uparrow}^{\lambda}(\theta_v) \rho_s^{\lambda}}{1 - \rho_s^{\lambda} S^{\lambda}} \tag{10}$$

where θ_s, θ_v, φ, and λ are the solar zenith angle, the observation zenith angle, the relative azimuth angle, and wavelength, respectively. ρ_0 is the path reflectance caused by the atmosphere, ρ_s is the surface reflectance, S is the spherical albedo of the atmosphere, $T_{\downarrow}$ and $T_{\uparrow}$ are the downward and upward transmittances of the atmosphere, respectively.

Theoretically, the simulated TOA reflectance corresponding to a certain aerosol content should be equal to the actually observed TOA reflectance. However, due to the existence of observation errors and surface reflectance errors, etc., there is no case that all simulated TOA reflectance is equal to the real TOA reflectance. Therefore, we obtain the optimal AOD retrieval result by minimizing the cost function J_L in (11). It is important to note that because the surface reflectance varies significantly with the observation geometry, while the ratio of surface reflectance of different bands does not vary essentially with the observation geometry [37], the ratio of surface reflectance of different bands instead of surface reflectance is used for AOD retrieval in this study

$$J_L(\tau) = \sum_{j=1}^{3}\left[\frac{\rho_{\mathrm{TOA}}^{j,\mathrm{meas}} - \rho_{\mathrm{TOA}}^{j,\mathrm{simu}}}{\rho_{\mathrm{TOA}}^{j,\mathrm{meas}}}\right]^2 \tag{11}$$

where j is the serial number of wavelength band, $\rho_{\mathrm{TOA}}^{j,\mathrm{meas}}$ is the real TOA reflectance, $\rho_{\mathrm{TOA}}^{j,\mathrm{simu}}$ is the simulated TOA reflectance.

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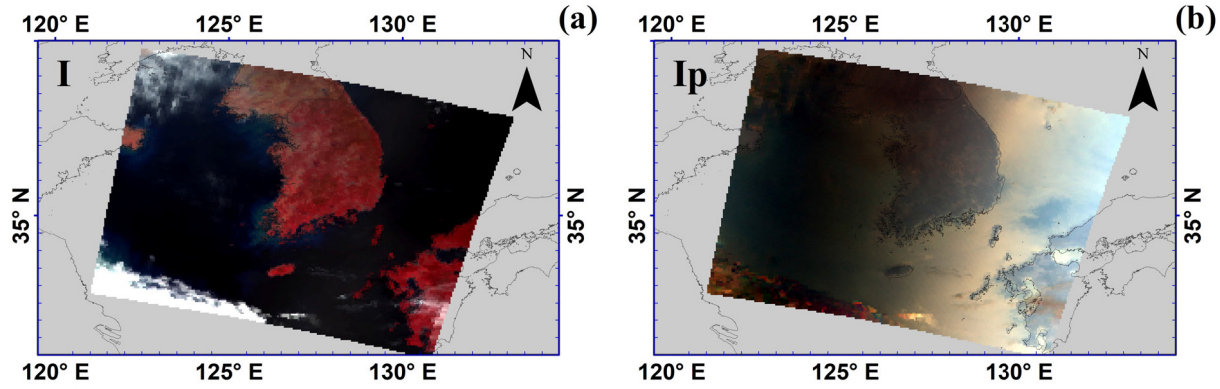


Fig. 5. PSAC false color map of Northeast Asia on April 11, 2021. (a) PSAC false color map synthesized by using nonpolarized data. (b) PSAC false color map synthesized by using polarization data.

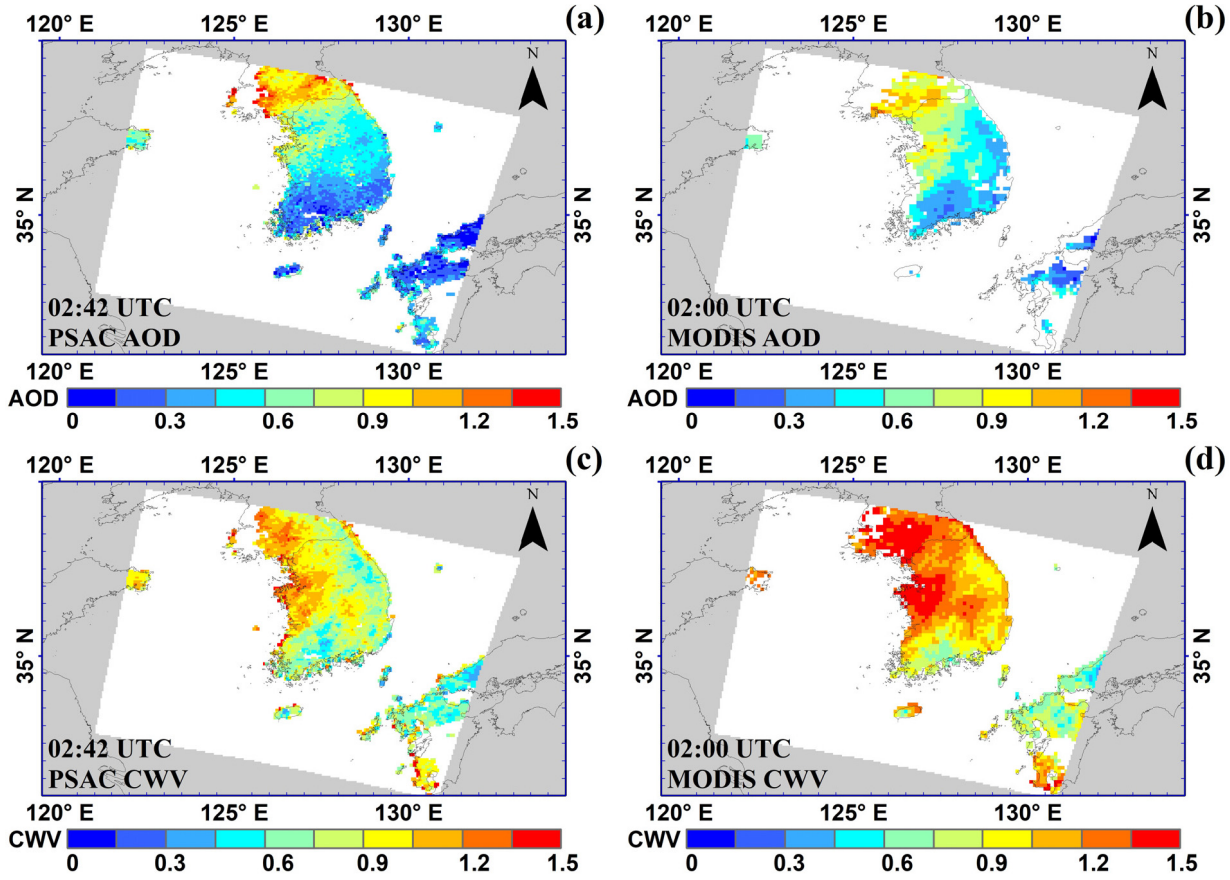


Fig. 6. Spatial distribution of PSAC data and corresponding MODIS data. (a) PSAC AOD data. (b) MODIS AOD data. (c) PSAC CWV data. (d) MODIS CWV data.

B. Results and Analysis

1) Spatial Distribution of AOD and CWV Retrieval Results: Examples of PSAC false color map and corresponding AOD and CWV retrieval results are shown in Figs. 5 and 6, respectively. In general, the spatial distribution trends of PSAC AOD data and MODIS AOD data are consistent. Although the spatial distribution trends of PSAC CWV data and MODIS CWV data are also generally consistent, the values of PSAC CWV data are significantly smaller than those of MODIS CWV data. This is because there is a significant systematic overestimation in the MODIS CWV data [38], while there is no significant overestimation in the PSAC CWV data. For example, the CWV value of MODIS CWV data at AERONET site Yonsei_University (37.564°N, 126.935°E) is equal to 1.17 cm, while the CWV value of its corresponding ground-based observation is 1.00 cm. In addition, the spatial distribution of the AOD and CWV data developed in this study

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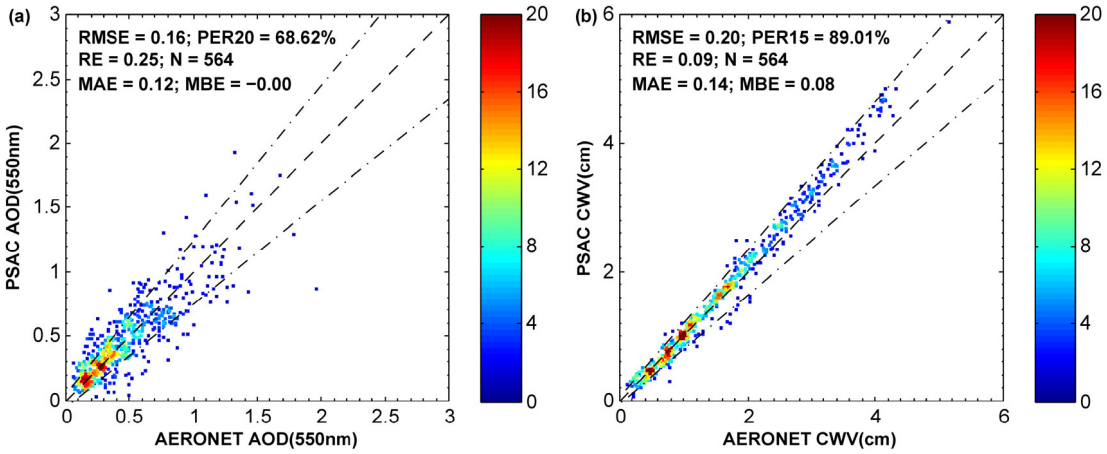


Fig. 7. Scatter plots of matching results between AERONET data and PSAC data from March 2021 to May 2021. The dotted line in the middle is the 1-1 line. The dotted lines on both sides of subfigure (a) are the envelope of PER20. The dotted lines on both sides of subfigure (b) are the envelope of PER15. The color of each point represents the density of the point where the point is located.

has good continuity and high spatial coverage, and there are basically no abnormal mutation retrieval results.

2) Overall Accuracy Assessment: In this section, the PSAC AOD and CWV data are validated by using ground-based data derived from AERONET [25], [39]. A total of five statistical parameters are used to evaluate the quality of PSAC AOD data, which contain the RMSE, MAE, mean bias error (MBE), relative error (RE), percentage of PSAC AOD data whose error falls within the expected error (EE) line $\pm(0.05 + 0.2 * \mathrm{AOD}_{\mathrm{AERONET}})$ (PER20) required for in-orbit test. For PSAC CWV data, the RMSE, MAE, MBE, RE and percentage of PSAC CWV data whose error falls within $\pm(0.05 + 0.15 * \mathrm{CWV}_{\mathrm{AERONET}})$ (PER15) are used for accuracy assessment of PSAC CWV data

$$\mathrm{MBE} = \sum_{i=1}^{N} \left(P_i - P_i'\right)/N \tag{12}$$

$$\mathrm{RE} = \sum_{i=1}^{N} \left|P_i - P_i'\right| / \sum_{i=1}^{N} P_i' \tag{13}$$

where N is the number of effective matching results, P_i is the value of PSAC data of the ith matching result, and P_i' is the value of AERONET data corresponding to P_i.

In this work, we use the mean value of AERONET observations within half an hour of the satellite observation time to match the spatial average of remote sensing retrieval results within the range of 50 km × 50 km centered on the AERONET site [40]. A valid match between AERONET data and remote sensing data needs to meet two requirements at the same time: 1) the number of valid remote sensing retrieval results within the range of 50 km × 50 km centered on the AERONET site is not less than 5 and 2) the number of AERONET observations within half an hour of the satellite observation time is not less than 2. The scatter plots of matching results between AERONET data and PSAC data are shown in Fig. 7. The RMSE, MAE, MBE, PER20, and RE of PSAC AOD data are 0.16, 0.12, 0, 68.62%, and 0.25, respectively. The MBE of the PSAC AOD dataset is approximately equal to zero. This means that there are basically no systematic deviations in the PSAC AOD data. In addition, the PER20 of the PSAC AOD data is greater than 68.27%, which indicates that the EE of the PSAC AOD data is better than $\pm(0.05 + 0.2 * \mathrm{AOD}_{\mathrm{AERONET}})$. The RMSE, MAE, MBE, PER15, and RE of PSAC CWV data are 0.20, 0.14, 0.08, 89.01%, and 0.09, respectively. Overall, there is a slight overestimation (MBE = 0.08) in the PSAC CWV data. The PER15 of the PSAC CWV data is greater than 68.27%, which indicates that the EE of the PSAC CWV data is better than $\pm(0.05 + 0.15 * \mathrm{CWV}_{\mathrm{AERONET}})$.

3) Time Series Results: To assess the temporal consistency between AERONET data and PSAC data, time series comparisons of AOD and CWV are performed at six AERONET sites, which are Beijing_RADI (40.005°N, 116.379°E), Gwangju_GIST (35.228°N, 126.843°E), Hankuk_UFS (37.339°N, 127.266°E), Jaipur (26.906°N, 75.806°E), Kanpur (26.513°N, 80.232°E), and Yonsei_University (37.564°N, 126.935°E). The time series comparison between PSAC AOD data and AERONET AOD data is shown in Fig. 8. Note that the AERONET data and remote sensing data in the time series comparison are matched according to the matching method mentioned above, so the AERONET and PSAC data in the time series comparison are in the same period of each day. Overall, PSAC AOD data and AERONET AOD data at all six sites show similar temporal trends. This indicates that the PSAC AOD data developed in this work can effectively characterize the temporal trend of AOD. However, there is a relatively large underestimation in the PSAC AOD data at Kanpur. The difference between the local actual aerosol model and the aerosol model (continental aerosol model [41]) used for PSAC AOD retrieval may be the main reason for the underestimation of PSAC AOD data at Kanpur. In future work, it is necessary to use AERONET data to build aerosol models that are closer to the actual situation to solve this problem. The time series comparison between AERONET CWV data and PSAC CWV data is shown in Fig. 9. Overall, PSAC CWV data and AERONET CWV data at all six sites are very close and show similar temporal trends. This indicates that PSAC CWV data developed in

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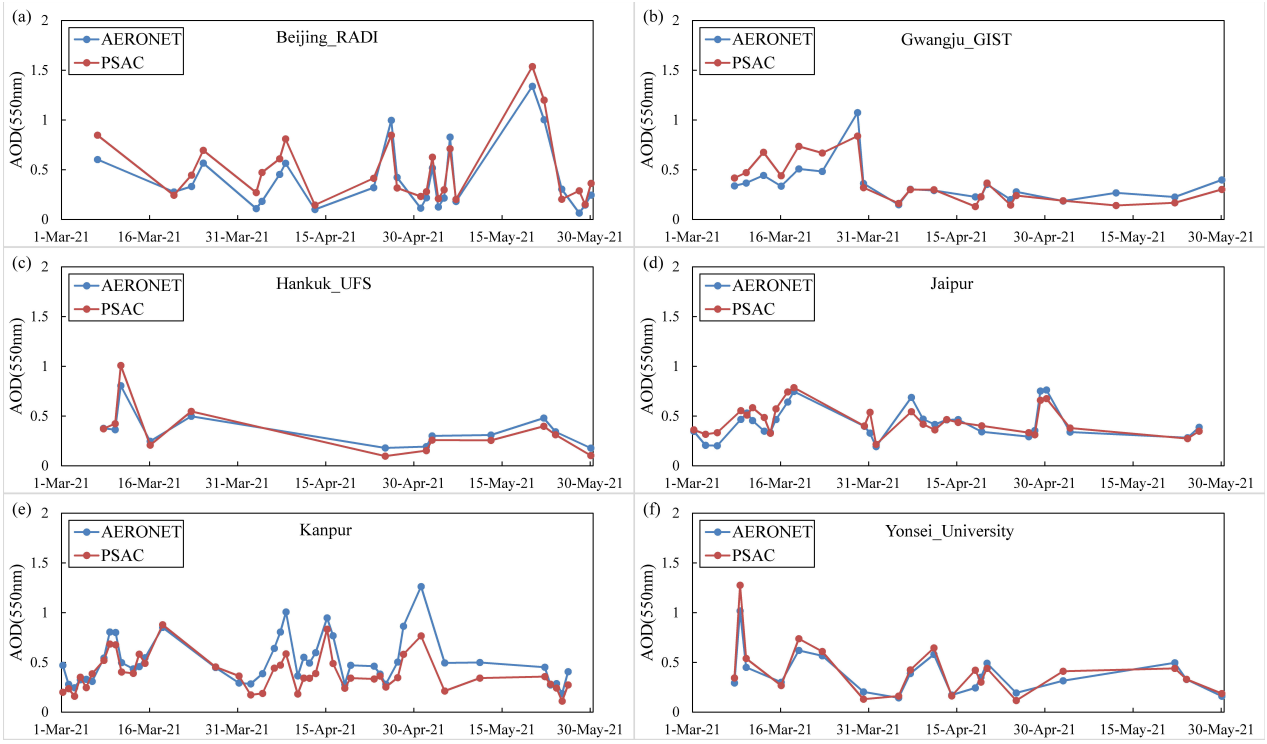


Fig. 8. Time series comparison between AERONET AOD data and PSAC AOD data. (a) Beijing_RADI site. (b) Gwangju_GIST site. (c) Hankuk_UFS site. (d) Jaipur site. (e) Kanpur site. (f) Yonsei_University site.

this work can effectively characterize the temporal trend of CWV.

IV. Synchronized Atmospheric Correction of CCD Image

A. Atmospheric Correction Method

In order to quantitatively and comprehensively use the CCD images of HJ-2 A/B satellites, the atmospheric correction needs to be carried out in advance, and thus the influence of atmospheric absorption and scattering can be removed to a certain extent. Based on the theories of atmospheric radiative transfer and adjacency effect of high-spatial-resolution image, the atmospheric blurring effect and adjacency effect are sequentially corrected in the cloud-free pixel of CCD image [42]–[44]. The steps of synchronized atmospheric correction are as follows.

Radiometric calibration of CCD image: As long as the Level 0 (L0) data has been obtained, we first convert the digital number (DN) of CCD image to the TOA radiance results by the radiometric calibration coefficients as

$$L = \frac{\mathrm{DN} - b}{a} \tag{14}$$

where L means the TOA radiance with the unit of $\mathrm{W} \cdot \mathrm{m}^{-2} \cdot \mathrm{sr}^{-1} \cdot \mu\mathrm{m}^{-1}$, a is the absolute calibration coefficient gain, and b is the offset. Correspondingly, the TOA reflectance can be calculated by

$$\rho_{\mathrm{TOA}} = \frac{\pi L}{E_0 \cos\theta_s} \tag{15}$$

where the E_0 is the extraterrestrial spectral solar irradiance.

Spatial matching of retrieved atmospheric parameters and CCD pixels: Due to the difference between pixel spatial resolution of PSAC (6 km) and CCD image (16 m), the spatial matching needs to be made by the latitude and longitude. That is to say, for each CCD pixel, the retrieval AOD and CWV results can be provided in the cloud-free pixel of PSAC data.

Atmospheric blurring effect correction: By assuming the surface is uniform Lambertian, the Lambertian surface reflectance ρ_s can be derived from (10) as (16), as shown at the bottom of the next page, where ρ_{TOA} is the TOA reflectance of CCD image, the results of path reflectance ρ_0, total transmission $T_{\downarrow}(\theta_s)T_{\uparrow}(\theta_v)$, hemispherical albedo S and gaseous transmission T_g are stored in advance in a lookup table simulated by 6SV to cover different observation geometry $(\theta_s, \theta_v, \varphi)$, AOD and CWV results [29]. In this way, the Lambertian and uniform assumed surface reflectance of CCD image can be calculated by the atmospheric blurring effect correction. However, the ρ_s here is still not the true surface reflectance (ρ_s^*) for the nonuniform surface case by considering the adjacency effect of 16 m CCD image [45]–[47].

Adjacency effect correction: For the high-spatial-resolution satellite image such as HJ-2 A/B CCD data, the adjacency effect of pixels also needs to be corrected, which is mainly caused by the atmospheric refraction of environmental contribution from the surrounding pixels to the atmospheric corrected pixel. After obtaining the uniform assumed surface reflectance ρ_s, the true surface reflectance ρ_s^* can be written as

$$\rho_s^* = \frac{\rho_s T_{\uparrow}(\theta_v) - \tilde{\rho}_s \left(T_{\uparrow}(\theta_v) - e^{-\tau/\cos\theta_v}\right)}{e^{-\tau/\cos\theta_v}} \tag{17}$$

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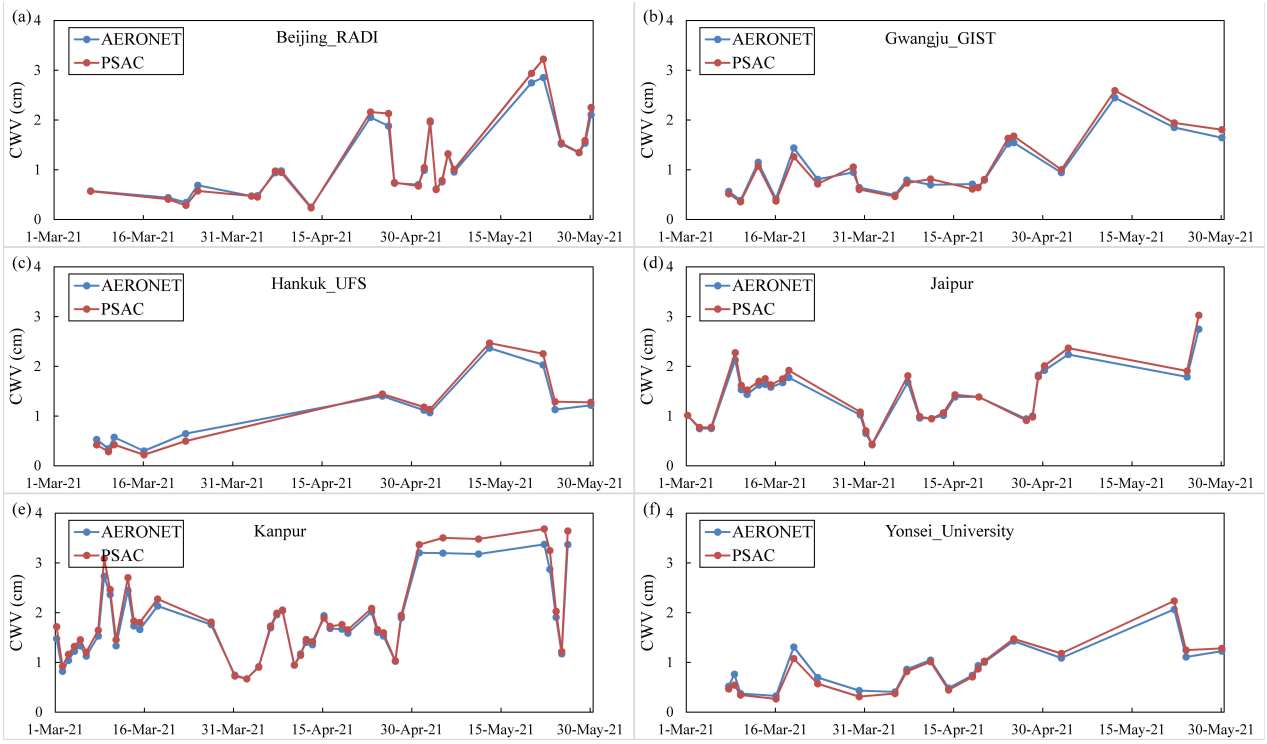


Fig. 9. Time series comparison between AERONET CWV data and PSAC CWV data. (a) Beijing_RADI site. (b) Gwangju_GIST site. (c) Hankuk_UFS site. (d) Jaipur site. (e) Kanpur site. (f) Yonsei_University site.

where τ means the atmospheric optical depth integrated by the AOD, Rayleigh optical depth, and gas optical depth, ρ_s^* means the true reflectance of target, $\tilde{\rho}_s$ means the contribution from the adjacent area centered on the pixel to be corrected, which is usually expressed by

$$\tilde{\rho}_s = \sum_{j=-n}^{n} \sum_{i=-n}^{n} f(r(i,j))\rho_s(i,j) \tag{18}$$

here $f(r)$ is the atmospheric point spread function, the $r(i,j)$ is the subfunction to represent the Euclidean distance between the pixels (i,j) and the center of the study zone [29], [48].

B. Atmospheric Correction Results

As an example, Fig. 10 shows the spatial matching between a scene of CCD image and the central location of synchronized PSAC pixels, which have been observed by the in-orbit HJ-2 satellite over the northeast region of Beijing in August 6, 2021. For the scene of CCD image, there are about 2000 × 2000 pixels with an area of 32 km × 32 km. Correspondingly, Fig. 11 illustrates the spatial distribution of AODs and CWVs retrieved from cloud-free PSAC data for synchronized atmospheric correction of CCD image. In the spatial coincidence area of CCD and PSAC measurements, based on the distance from the position of CCD pixel to the center position of PSAC pixel, the nearest neighbor method is used for spatial matching between CCD and PSAC. Obviously, the scene of CCD image can be roughly divided into 5 × 5 subregions in this example, and each subregion of CCD can be matched with a set of AOD and CWV result from PSAC.

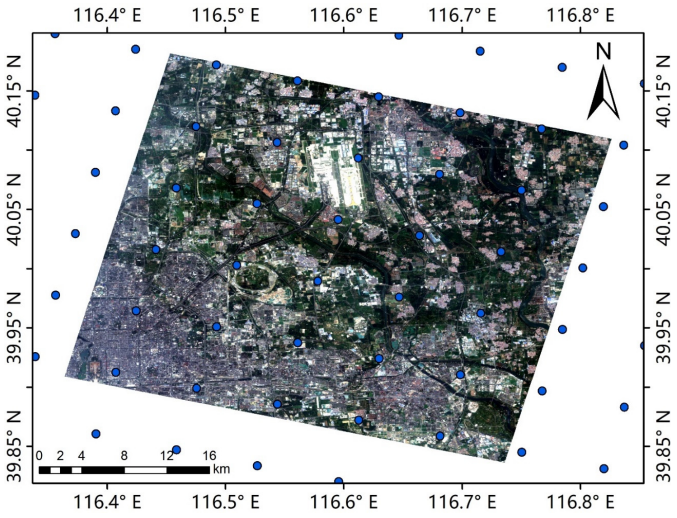


Fig. 10. Spatial matching between a scene of CCD image and the central location of corresponding PSAC's pixels. The blue dot represents the center latitude and longitude location of each PSAC pixel.

$$\rho_s = \frac{\rho_{\mathrm{TOA}}(\theta_s,\theta_v,\varphi) - \rho_0(\theta_s,\theta_v,\varphi)T_g(\theta_s,\theta_v)}{\left[\rho_{\mathrm{TOA}}(\theta_s,\theta_v,\varphi) - \rho_0(\theta_s,\theta_v,\varphi)T_g(\theta_s,\theta_v)\right]S + T_{\downarrow}(\theta_s)T_{\uparrow}(\theta_v)T_g(\theta_s,\theta_v)} \tag{16}$$

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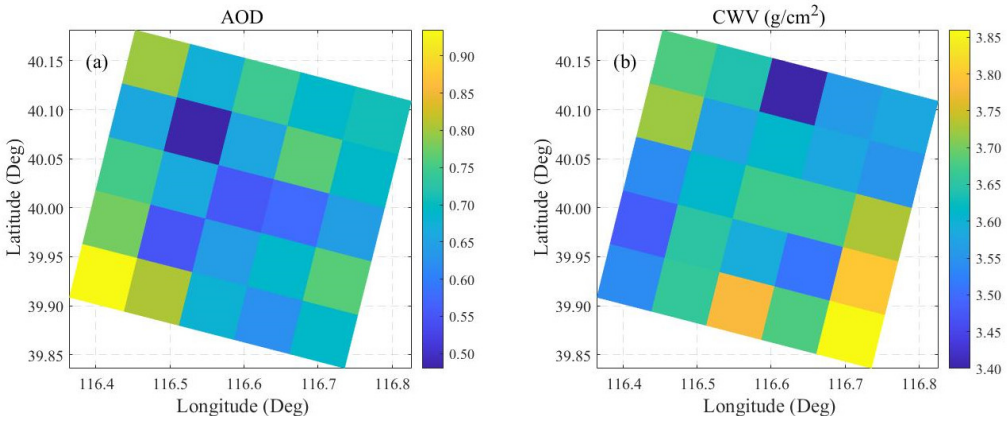


Fig. 11. Corresponding spatial distribution of (Left) AODs and (Right) CWVs retrieved from PSAC for synchronized atmospheric correction of CCD image by the spatial matching. (a) AOD retrieval result. (b) CWV retrieval result.

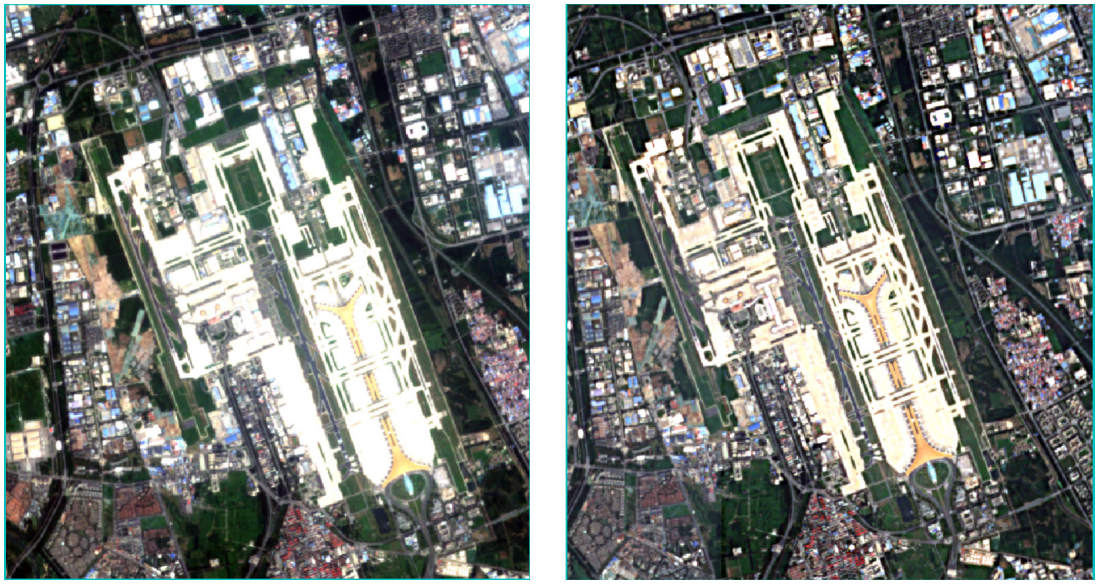


Fig. 12. True color composite image before and after correction [(Left): original image ρ_{TOA}, (Right): after blurring effect and adjacency effect correction ρ_s^*].

In this way, the nearest-neighbor results of AOD and CWV can be matched for cloud-free pixels of CCD for atmospheric blurring effect correction, and then the adjacency effect of CCD image can further be corrected.

Fig. 12 shows the original true color composite image (ρ_{TOA}) and the corrected image (ρ_s^*) after blurring effect plus adjacency affect correction. Compared to the original image, we can find that the image texture detail and sharpness have a significant improvement after atmospheric blurring effect and adjacency effect correction. Obviously, it would be great if the multispectral surface reflectance results of CCD image after atmospheric correction can be validated against the simultaneous ground measurements. However, due to the lack of the *in situ* surface reflectance data with the transiting observation of HJ-2 A/B satellites, the synchronized validation of atmospheric correction results is not yet available in this article.

In order to quantitatively evaluate the effect of atmospheric correction, four evaluation parameters are used, including clarity, detail signal energy, edge energy, and contrast. Fig. 13 illustrates the values of these evaluation parameters in the format of histograms for the image of ρ_{TOA}, ρ_s, and ρ_s^*, respectively. Correspondingly, Table III lists the quantitative promotion of atmospheric correction by the results of ρ_s versus ρ_{TOA} and ρ_s^* versus ρ_{TOA}. Taking the promotion results of "$\rho_s/\rho_{\mathrm{TOA}}$" as an example, the percentage ratio of 100% means no promotion by ρ_s versus ρ_{TOA}, and the corresponding evaluation parameter's value of ρ_s is equal to the evaluation value of ρ_{TOA}. Generally, for the atmospheric correction at five wavelength bands of CCD image, the shorter of the wavelength in VIS-NIR, the more obvious of the correction effect, and the larger value of evaluation parameters, that is because shorter visible band (e.g., 490 nm) is more susceptible to the influence of the atmosphere by Rayleigh scattering and aerosol scattering than the NIR wavelength band (e.g., 865 nm).

V. Discussion

In this study, the ratios of surface reflectance of different bands are necessary for aerosol retrieval. First, the surface reflectance of different bands of PSAC is calculated by NMF

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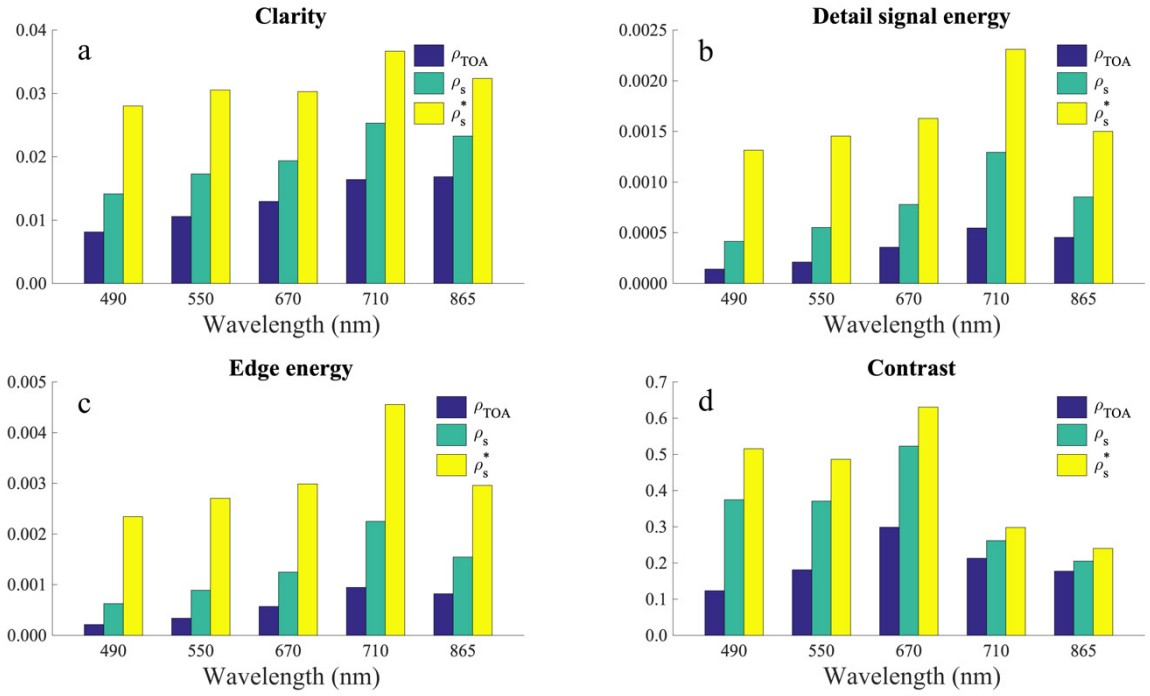


Fig. 13. Quantitative evaluation of the atmospheric blurring effect and adjacency effect correction by (a) parameters of clarity, (b) detail signal energy, (c) edge energy, and (d) contrast, respectively.

TABLE III
COMPARISON OF IMAGE QUANTITATIVE IMPROVEMENT BY ATMOSPHERIC CORRECTION

CCD Band	Clarity improvement		Detail signal energy improvement		Edge energy improvement		Contrast improvement	
	ρ_s/ρ_{TOA}	ρ_s^*/ρ_{TOA}	ρ_s/ρ_{TOA}	ρ_s^*/ρ_{TOA}	ρ_s/ρ_{TOA}	ρ_s^*/ρ_{TOA}	ρ_s/ρ_{TOA}	ρ_s^*/ρ_{TOA}
490 nm	174%	345%	293%	924%	292%	1094%	303%	417%
550 nm	164%	289%	261%	686%	261%	794%	205%	268%
670 nm	149%	234%	219%	456%	219%	524%	175%	211%
710 nm	154%	224%	237%	423%	238%	482%	123%	140%
865 nm	138%	192%	188%	330%	188%	360%	116%	135%

method using MODIS surface reflectance data [34], and then the ratios of surface reflectance of different bands of PSAC are calculated using the surface reflectance of these bands. The accuracy of these ratios is positively correlated with that of AOD retrieval results. Although we have selected the spectral reflectance curves of as many typical features as possible when constructing the spectral vectors, it is impossible to include all spectral reflectance curves of the features. In addition, the error of MODIS surface reflectance data has a negative impact on the accuracy of PSAC surface reflectance data. The above-mentioned reasons all have a negative impact on the accuracy of the ratio we construct. In addition to the method used in this study, we can use the background AOD data to perform atmospheric correction of the subdark pixels over a month to obtain the surface reflectance of different bands of PSAC, and then obtain the ratio of the surface reflectance of different channels based on these surface reflectance results. The method requires long time series of data to construct the ratio of surface reflectance of different channels [31]. The PSAC is still in the in-orbit testing stage and the acquired data are still relatively small, so we currently do not use this method to construct the ratio of the surface reflectance of different channels.

Since the spatial resolution of PSAC data is 6 km × 6 km and the spatial resolution of CCD data is about 16 m × 16 m, one pixel of PSAC data corresponds to hundreds of thousands of pixels of CCD data. Therefore, if a pixel of PSAC data is discriminated as a cloudy pixel because part of the area corresponding the pixel is covered by clouds, it can lead to some pixels of CCD data corresponding to the pixel cannot be corrected due to the lack of AOD and CWV retrieval results. However, due to the uniformity of the spatial distribution of aerosol and water vapor in the range of tens of kilometers [40], the aerosol and water vapor retrieval result of adjacent pixels of the PSAC can be used to perform atmospheric correction on the CCD data to avoid this situation. However, if there is no valid retrieval results over the adjacent pixels due to the existence of clouds, this method is invalid. In addition, if a

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PSAC pixel contains a small cloud that is not identified by the cloud identification algorithm, it may cause the retrieval results of AOD and CWV to deviate significantly from the true values. In fact, this situation is unavoidable due to the low spatial resolution of PSAC data. In order to reduce the occurrence of this situation, it is necessary to improve the spatial resolution of PSAC data.

VI. Conclusion

In order to test the in-orbit performance of PSAC onboard HJ-2 A/B satellites, the "day-1" algorithms have been developed and briefly introduced in this study, including the cloud mask algorithm, water vapor, and aerosol retrieval algorithms. Based on the in-orbit measurements of PSAC mounted on HJ-2 A/B satellites from March to May in 2021, the AOD and CWV products are retrieved from the cloud-free pixels, and the corresponding retrieval accuracies are evaluated by against the ground-based AERONET results. The preliminary validation results show that the CWV retrieval accuracy is better than the EE of $\pm(0.05+0.15*\mathrm{CWV_{AERONET}})$, while the AOD retrieval accuracy is better than the EE of $\pm(0.05+0.20*\mathrm{AOD_{AERONET}})$ required for in-orbit test. In addition, both of the CWV and AOD retrieval results have broad spatial coverage, and can effectively characterize the change trend of these two key atmospheric parameters.

After that, to further eliminate the negative impact of the atmosphere on CCD/HJ-2 multispectral data and expand its application range, synchronized atmospheric correction of CCD image has been tested with the spatial-matched AODs and CWVs retrieved from PSAC, in which both of atmospheric blurring effect correction and adjacency effect correction are considered. Compared with the original 16 m CCD image, the corresponding texture detail and quality of the atmospheric corrected multispectral image can be significantly improved. The clarity, detail signal energy, edge energy, and contrast of visible CCD data after atmospheric are improved by at least 111%, and those of NIR CCD data after atmospheric correction are improved by at least 35%. To sum up, the results of in-orbit test in article preliminarily show the good application potential of PSAC in atmospheric monitoring and atmospheric correction.

Appendix

In this study, there are four evaluation parameters are selected for the image quantitative evaluation of atmospheric correction, including clarity, detail signal energy, edge energy, and contrast.

A. Clarity of Image

The clarity of image mainly indicates the clarity of the image texture, which can be expressed by the sum of the gray change speeds between adjacent pixels in x and y directions as follows:

$$\sigma_l(x,y)=\frac{1}{m\times n}\sum_{y=1}^{m-1}\sum_{x=1}^{n-1}(|\,f(x,y)-f(x,y+1)\,|+|\,f(x,y)-f(x+1,y)\,|). \quad \text{(A1)}$$

TABLE IV
Four Common Edge Extraction Operators

No.	E_1	E_2
1	$\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}$	$\begin{bmatrix}0 & 1\\ -1 & 0\end{bmatrix}$
2	$\begin{bmatrix}\frac{1}{3} & \frac{1}{3} & 0\\ \frac{1}{3} & 0 & -\frac{1}{3}\\ 0 & -\frac{1}{3} & -\frac{1}{3}\end{bmatrix}$	$\begin{bmatrix}0 & \frac{1}{3} & \frac{1}{3}\\ -\frac{1}{3} & 0 & \frac{1}{3}\\ -\frac{1}{3} & -\frac{1}{3} & 0\end{bmatrix}$
3	$\begin{bmatrix}\frac{1}{2} & \frac{1}{4} & 0\\ \frac{1}{4} & 0 & -\frac{1}{4}\\ 0 & -\frac{1}{4} & -\frac{1}{2}\end{bmatrix}$	$\begin{bmatrix}0 & \frac{1}{4} & \frac{1}{2}\\ -\frac{1}{4} & 0 & \frac{1}{4}\\ -\frac{1}{2} & -\frac{1}{4} & 0\end{bmatrix}$
4	$\begin{bmatrix}\frac{1}{6} & -\frac{1}{6} & -\frac{1}{6}\\ -\frac{1}{6} & \frac{4}{6} & -\frac{1}{6}\\ -\frac{1}{6} & -\frac{1}{6} & \frac{1}{6}\end{bmatrix}$	$\begin{bmatrix}-\frac{1}{6} & -\frac{1}{6} & \frac{1}{6}\\ -\frac{1}{6} & \frac{4}{6} & -\frac{1}{6}\\ \frac{1}{6} & -\frac{1}{6} & -\frac{1}{6}\end{bmatrix}$

The result of image clarity is equivalent to the sum of the gray difference between each pixel and the surrounding pixels. The more the edge of the image, the greater the change of regional gray scale, and the greater the value of clarity.

B. Energy of the Image Detail Signal

The energy of the image detail signal is the mean value of the variance of the local area of the image, which reflects the detail signal, image texture and edge of the image, etc., and has a strong correlation with the image variance. By taking an area of $(2M+1)\times(2M+1)$ pixel size on the image, the mean gray value of this area can be written as

$$m_f(x,y)=\frac{1}{(2M+1)^2}\sum_{i=-M}^{M}\sum_{j=-M}^{M}f(x+i,y+j) \quad \text{(A2)}$$

and the variance of this area is

$$\sigma_f^2(x,y)=\frac{1}{(2M+1)^2}\sum_{i=-M}^{M}\sum_{j=-M}^{M}\left[f(x+i,y+i)-m_f(x,y)\right]^2. \quad \text{(A3)}$$

Thus, we have the expression of image detail signal energy as

$$\sigma_f^2=\frac{1}{m\times n}\sum_{x=1}^{m}\sum_{y=1}^{n}\sigma_f^2(x,y). \quad \text{(A4)}$$

When the image is less affected by noise, the larger the value, the finer the texture of the image and the clearer the image.

C. Edge Energy of Image

The edge is a component of the texture and shape of the image, and it has the directional characteristics. Correspondingly, the edge energy of image can be obtained by convolving

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the edge operator as

$$e(x, y) = E_1(f(x, y)) + E_2(f(x, y)) \tag{A5}$$

where E_1 and E_2, respectively, mean the edge operators in 45° and 135° directions usually used, and four common edge extraction operators are listed in Table IV. Sequentially, we have the mean energy at the edge of the image

$$\sigma_e^2 = \frac{1}{m \times n} \sum_{x=1}^{m} \sum_{y=1}^{n} e^2(x, y). \tag{A6}$$

The edge energy can highlight the severity of the image grayscale changes, and the large value represents that the image grayscale changes obviously and the texture is complex.

D. Contrast of Image

Contrast refers to the measurement of different brightness levels between light and dark areas in an image, the larger the difference range, the greater the contrast. In addition, the smaller the difference range, the smaller the contrast. Therefore, a better contrast can make the image display vivid and rich colors, which can be expressed as

$$\sigma_c(x, y) = \frac{\sqrt{\frac{1}{m \times n} \sum_{y=1}^{m} \sum_{x=1}^{n} f^2(x, y) - \left(\frac{1}{m \times n} \sum_{y=1}^{m} \sum_{x=1}^{n} f(x, y)\right)^2}}{\frac{1}{m \times n} \sum_{y=1}^{m-1} \sum_{x=1}^{n-1} f(x, y)} \tag{A7}$$

where m and n are the width and height of the image, respectively, and $f(x, y)$ is the value of the pixel point in the x row and y column.

ACKNOWLEDGMENT

The MODIS data used in this work are provided by National Aeronautics and Space Administration (NASA). The authors thank them for their data support for this study. The ground-based aerosol optical depth (AOD) and columnar water vapor (CWV) data used to validate the remote sensing data are obtained from Aerosol Robotic Network (AERONET) sites. We would also like to thank the PI(s), Co-I(s), and their staff for their efforts in establishing and maintaining the AERONET sites, especially those of Beijing_RADI, Gwangju_GIST, Hankuk_UFS, Jaipur, Kanpur, and Yonsei_University. The authors would like to thank Xinkai Hu from Hainan University for the helpful work on atmospheric correction data processing during his short-term guest study in Aerospace Information Research Institute, Chinese Academy of Sciences, as well as the help from Xingyun He for revising the grammar of this article. In addition, Weizhen Hou would like to thank the UNL-VRTM code support from the developers of Unified Linearized Vector Radiative Transfer Model (https://unl-vrtm.org/).

REFERENCES

[1] R. C. Levy *et al.*, "The Collection 6 MODIS aerosol products over land and ocean," *Atmos. Meas. Techn.*, vol. 6, no. 11, pp. 2989–3034, 2013.

[2] A. Mitra, L. Di Girolamo, Y. L. Hong, Y. Z. Zhan, and K. J. Mueller, "Assessment and error analysis of Terra-MODIS and MISR cloud-top heights through comparison with ISS-CATS lidar," *J. Geophys. Res., Atmos.*, vol. 126, no. 9, May 2021, Art. no. e2020JD034281.

[3] N. F. Stevens, H. Garbeil, and P. J. Mouginis-Mark, "NASA EOS Terra ASTER: Volcanic topographic mapping and capability," *Remote Sens. Environ.*, vol. 90, no. 3, pp. 405–414, Apr. 2004.

[4] D. P. Edwards, G. Petron, P. C. Novelli, L. K. Emmons, J. C. Gille, and J. R. Drummond, "Southern hemisphere carbon monoxide interannual variability observed by Terra/measurement of pollution in the troposphere (MOPITT)," *J. Geophys. Res. Atmos.*, vol. 111, no. D16, Aug. 2006, Art. no. D16303.

[5] S. L. Bevan, P. R. J. North, S. O. Los, and W. M. F. Grey, "A global dataset of atmospheric aerosol optical depth and surface reflectance from AATSR," *Remote Sens. Environ.*, vol. 116, pp. 199–210, Jan. 2012.

[6] N. Gobron *et al.*, "Evaluation of the MERIS/ENVISAT FAPAR product," *Adv. Space Res.*, vol. 39, no. 1, pp. 105–115, 2007.

[7] H. Lin, J. Chen, Z. Pei, S. Zhang, and X. Hu, "Monitoring sugarcane growth using ENVISAT ASAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 8, pp. 2572–2580, Aug. 2009.

[8] E. Kyrola *et al.*, "GOMOS on ENVISAT: An overview," in *Climate Change Processes in Stratosphere, Earth-Atmosphere-Ocean Systems, Oceanographic Processes From Satellite Data* (Advances in Space Research), vol. 33, P. Schlussel, R. Stuhlmann, J. W. Campbell, and C. Erickson, Eds. Amsterdam, The Netherlands: Elsevier, 2004, pp. 1020–1028, doi: 10.1016/S0273-177(03)00590-8.

[9] W. Chen *et al.*, "Estimating $PM_{2.5}$ with high-resolution 1-km AOD data and an improved machine learning model over Shenzhen, China," *Sci. Total Environ.*, vol. 746, Dec. 2020, Art. no. 141093.

[10] M. Shrivastava *et al.*, "Recent advances in understanding secondary organic aerosol: Implications for global climate forcing," *Rev. Geophys.*, vol. 55, no. 2, pp. 509–559, Jun. 2017.

[11] I. M. Held and B. J. Soden, "Water vapor feedback and global warming," *Annu. Rev. Energy Environ.*, vol. 25, no. 1, pp. 441–475, Nov. 2000.

[12] D. Tanré *et al.*, "Remote sensing of aerosols by using polarized, directional and spectral measurements within the A-train: The PARASOL mission," *Atmos. Meas. Tech.*, vol. 4, pp. 1383–1395, Apr. 2011.

[13] L. Liu, C. Weng, S. Li, L. Husi, S. Hu, and P. Dong, "Passive remote sensing of ice cloud properties at terahertz wavelengths based on genetic algorithm," *Remote Sens.*, vol. 13, no. 4, p. 735, Feb. 2021.

[14] S. Persh *et al.*, "Ground performance measurements of the glory aerosol polarimetry sensor," *Proc. SPIE*, vol. 7807, Aug. 2010, Art. no. 780703.

[15] F. M. Bréon *et al.*, "Scientific results from the polarization and directionality of the Earth's reflectances (POLDER)," in *Earth's Atmosphere, Ocean Surface Studies* (Advances in Space Research), vol. 30, J. L. Fellous, J. F. LeMarshall, B. J. Choudhury, M. Menenti, L. J. Paxton, and S. P. Gupta, Eds. Amsterdam, The Netherlands. Elsevier, 2002, pp. 2383–2386, doi: 10.1016/S0273-1177(02)80282-4.

[16] Z. Li *et al.*, "Directional polarimetric camera (DPC): Monitoring aerosol spectral optical properties over land from satellite observation," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 218, pp. 21–37, Oct. 2018.

[17] B. Fougnie *et al.*, "The multi-viewing multi-channel multi-polarisation imager—Overview of the 3MI polarimetric mission for aerosol and cloud characterization," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 219, pp. 23–32, Nov. 2018.

[18] W. Hou, "Preliminary study on the multispectral measurement simulations of polarized scanning atmospheric corrector image," *Proc. SPIE*, vol. 11566, Nov. 2020, Art. no. 115660E.

[19] M. I. Mishchenko *et al.*, "Accurate monitoring of terrestrial aerosols and total solar irradiance: Introducing the glory mission," *Bull. Amer. Meteorol. Soc.*, vol. 88, no. 5, pp. 677–692, May 2007.

[20] O. Dubovik *et al.*, "Polarimetric remote sensing of atmospheric aerosols: Instruments, methodologies, results, and perspectives," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 224, pp. 474–511, Feb. 2019.

[21] X. F. Lei *et al.*, "Integration model of POSP measurement spatial response function," *Opt. Exp.*, vol. 28, no. 17, pp. 25480–25490, Aug. 2020.

[22] J. E. Hansen and L. D. Travis, "Light scattering in planetary atmospheres," *Space Sci. Rev.*, vol. 16, no. 4, pp. 527–610, 1974.

[23] M. Moradizadeh, M. Momeni, and M. R. Saradjian, "Estimation and validation of atmospheric water vapor content using a MODIS NIR band ratio technique based on AIRS water vapor products," *Arabian J. Geosci.*, vol. 7, no. 5, pp. 1891–1897, May 2014.

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[24] J. A. Sobrino, J. El Kharraz, and Z.-L. Li, "Surface temperature and water vapour retrieval from MODIS data," *Int. J. Remote Sens.*, vol. 24, no. 24, pp. 5161–5182, Dec. 2003.
[25] D. M. Giles *et al.*, "Advancements in the aerosol robotic network (AERONET) version 3 database-automated near-real-time quality control algorithm with improved cloud screening for sun photometer aerosol optical depth (AOD) measurements," *Atmos. Meas. Techn.*, vol. 12, no. 1, pp. 169–209, Jan. 2019.
[26] Y. Xue, X. He, H. Xu, J. Guang, J. Guo, and L. Mei, "China Collection 2.0: The aerosol optical depth dataset from the synergetic retrieval of aerosol properties algorithm," *Atmos. Environ.*, vol. 95, pp. 45–58, Oct. 2014.
[27] F. K. Yang *et al.*, "Preliminary investigation of a new AHI aerosol optical depth (AOD) retrieval algorithm and evaluation with multiple source AOD measurements in China," *Remote Sens.*, vol. 10, no. 5, May 2018, Art. no. 748.
[28] Y. Xie *et al.*, "Columnar water vapor retrieval by using data from the polarized scanning atmospheric corrector (PSAC) onboard HJ-2 A/B satellites," *Remote Sens.*, vol. 14, no. 6, p. 1376, Mar. 2022.
[29] E. F. Vermote, D. Tanre, J. L. Deuze, M. Herman, and J. J. Morcette, "Second simulation of the satellite signal in the solar spectrum, 6S: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 35, no. 3, pp. 675–686, May 1997.
[30] Y. Xie *et al.*, "Aerosol optical depth retrieval over South Asia using FY-4A/AGRI data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.
[31] K. R. Knapp, R. Frouin, S. Kondragunta, and A. Prados, "Toward aerosol optical depth retrievals over land from GOES visible radiances: Determining surface reflectance," *Int. J. Remote Sens.*, vol. 26, pp. 4097–4116, Sep. 2005.
[32] A. I. Prados, S. Kondragunta, P. Ciren, and K. R. Knapp, "GOES aerosol/smoke product (GASP) over North America: Comparisons to AERONET and MODIS observations," *J. Geophys. Res.*, vol. 112, no. D15, Aug. 2007, Art. no. D15201.
[33] E. F. Vermote and A. Vermeulen, "Atmospheric correction algorithm: Spectral reflectances (MOD09)," *ATBD Version*, vol. 4, pp. 1–107, Apr. 1999.
[34] W. Hou, "Hyperspectral surface reflectance reconstruction based on non-negative matrix factorization and multispectral results," *Proc. SPIE*, vol. 12064, p. 25, Nov. 2021.
[35] W. Hou, "Study on the spectral reconstruction of typical surface types based on spectral library and principal component analysis," *Proc. SPIE*, vol. 11023, p. 170, Mar. 2019.
[36] Y. J. Kaufman *et al.*, "Passive remote sensing of tropospheric aerosol and atmospheric correction for the aerosol effect," *J. Geophys. Res., Atmos.*, vol. 102, no. D14, pp. 16815–16830, Jul. 1997.
[37] Y. Xie *et al.*, "Deriving a global and hourly data set of aerosol optical depth over land using data from four geostationary satellites: GOES-16, MSG-1, MSG-4, and Himawari-8," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1538–1549, Mar. 2020.
[38] Y. Xie *et al.*, "Validation of FY-3D MERSI-2 precipitable water vapor (PWV) datasets using ground-based PWV data from AERONET," *Remote Sens.*, vol. 13, no. 16, p. 3246, Aug. 2021.
[39] B. N. Holben *et al.*, "AERONET—A federated instrument network and data archive for aerosol characterization," *Remote Sens. Environ.*, vol. 66, no. 1, pp. 1–16, Oct. 1998.
[40] C. Ichoku, "A spatio-temporal approach for global validation and analysis of MODIS aerosol products," *Geophys. Res. Lett.*, vol. 29, no. 12, p. 1616, Jun. 2002.
[41] J. Lenoble and C. Brogniez, "A comparative review of radiation aerosol models," *Beitraege Phys. Atmos.*, vol. 57, pp. 1–20, Feb. 1984.
[42] H. Li *et al.*, "Atmospheric correction of geostationary satellite ocean color data under high solar zenith angles in open oceans," *Remote Sens. Environ.*, vol. 249, Nov. 2020, Art. no. 112022.
[43] Y. J. Kaufman and C. Sendra, "Algorithm for automatic atmospheric corrections to visible and near-IR satellite imagery," *Int. J. Remote Sens.*, vol. 9, no. 8 pp. 1357–1381, Aug. 1988.
[44] T. Wang *et al.*, "An adaptive atmospheric correction algorithm for the effective adjacency effect correction of submeter-scale spatial resolution optical satellite images: Application to a WorldView-3 panchromatic image," *Remote Sens. Environ.*, vol. 259, Jun. 2021, Art. no. 112412.
[45] D. R. Thompson *et al.*, "A unified approach to estimate land and water reflectances with uncertainties for coastal imaging spectroscopy," *Remote Sens. Environ.*, vol. 231, Sep. 2019, Art. no. 111198.
[46] Y. Pan, F. Shen, and W. Verhoef, "An improved spectral optimization algorithm for atmospheric correction over turbid coastal waters: A case study from the Changjiang (Yangtze) estuary and the adjacent coast," *Remote Sens. Environ.*, vol. 191, pp. 197–214, Mar. 2017.
[47] M. A. Warren *et al.*, "Assessment of atmospheric correction algorithms for the Sentinel-2A multispectral imager over coastal and inland waters," *Remote Sens. Environ.*, vol. 225, pp. 267–289, May 2019.
[48] L. C. Sanders, J. R. Schott, and R. Raqueño, "A VNIR/SWIR atmospheric correction algorithm for hyperspectral imagery with adjacency effect," *Remote Sens. Environ.*, vol. 78, no. 3, pp. 252–263, Dec. 2001.



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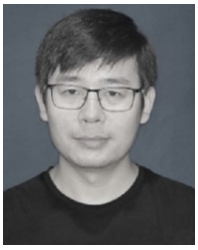
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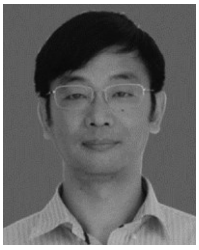
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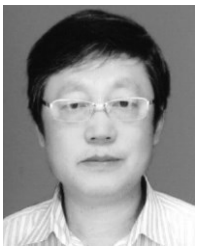
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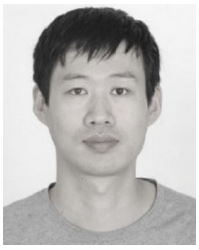
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